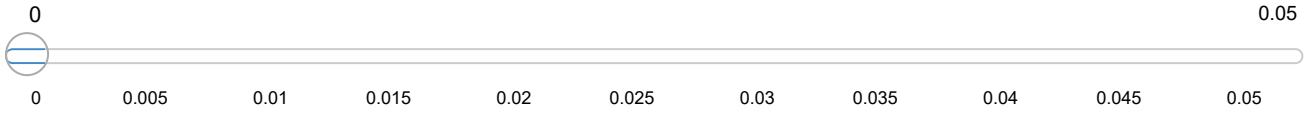
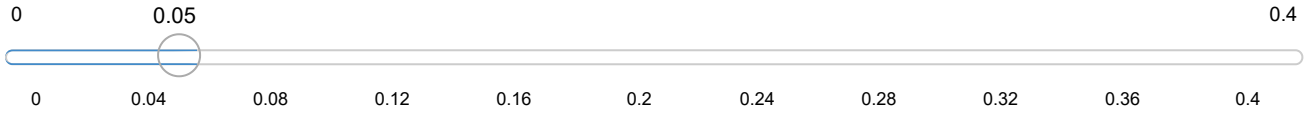
Evolutionary Rescue Simulation

Starting frequency of mutants p0:



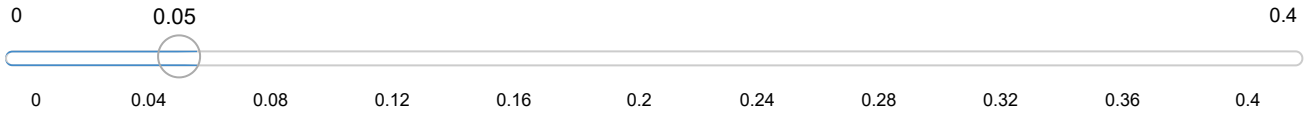
This parameter determines the standing genetic variation. If it is 0, only new mutations can rescue the population

Rate of population decline r:



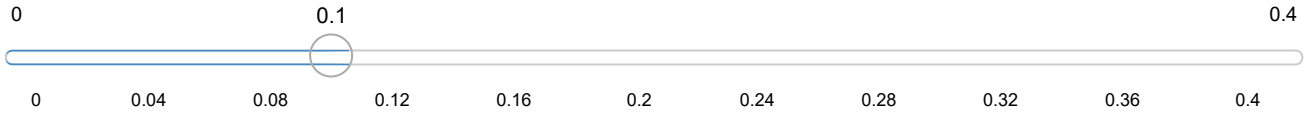
This parameter determines how fast the population initially declines

Rate of migration m:



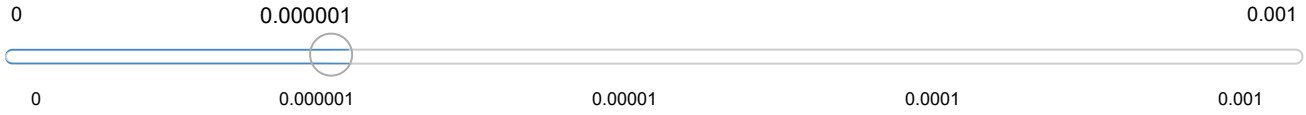
This parameter determines how many individuals migrate into the new Habitat

Selective advantage of mutants s:



Advantage must be greater than decline rate (s>r) to lift fitness of mutants above 1.

Mutation probability u:



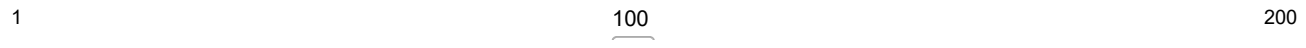
This is the probability at which a beneficial mutation occurs per new offspring

Starting population size N0:



This is the population size at time 0

Number of generations to simulate:



1 21 41 61 81 101 121 141 161 181 200

This determines how long the simulation is run

Number of populations to simulate:

1 5 100

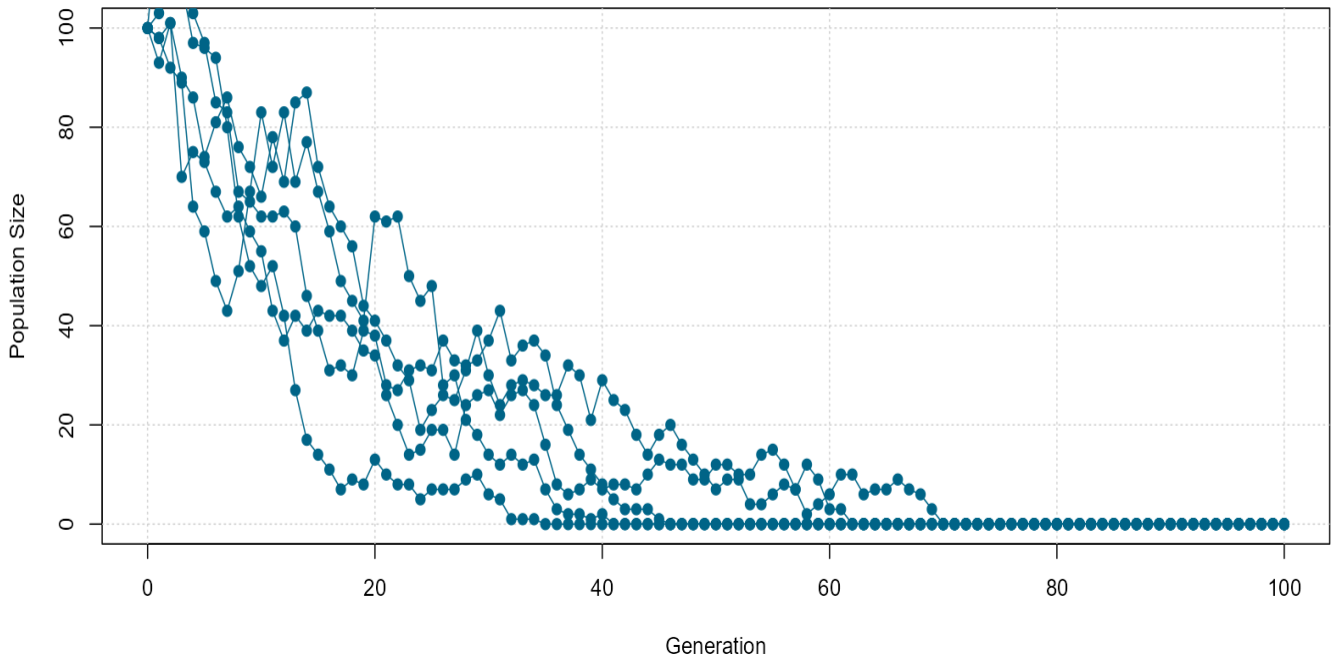
1 11 21 31 41 51 61 71 81 91 100

This determines how many times the simulation is run, i.e., how many trajectories will appear in the plot

Recalculate

Use this button to rerun the simulation without changing the parameters

Simulated population sizes over time



Population rebound: 0/5 (0%)
Expected rescue probability: 0.02 %